Fiscal Year:	FY 2010	**Task Last Updated:**	FY 02/01/2010
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Levitation Observation of Dendrite Evolution in Steel Ternary Alloy Rapid Solidification (LODESTARS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	douglas.matson@tufts.edu	**Fax:**	FY (617) 627-3058
PI Organization Type:	UNIVERSITY	**Phone:**	(617) 627-3239
Organization Name:	Tufts University		
PI Address 1:	Mechanical Engineering		
PI Address 2:	025 Anderson Hall, 200 College Ave		
PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	04/03/2008	**End Date:**	04/02/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Szofran, Frank	**Contact Phone:**	256-544-7777
Contact Email:	frank.szofran@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hyers, Robert (University of Massachusetts, Amherst)		
Grant/Contract No.:	NNX08AL21G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Undercooling in containerless processing has shown promise in producing metastable phases and materials that exhibit improved chemical homogeneity and ultra-fine grain sizes. Stainless steel alloys require special handling in order to achieve a significant undercooling range. In particular, chromium is particularly susceptible to contamination through reaction with crucible material. When attempting to perform rapid solidification tests, the formation of active heterogeneous nucleation sites limits the undercooling that may be achieved. By using a containerless technique, this contamination is minimized. The focus of this program is to study of the effect of microgravity on metastable solidification. By observing changes in the mechanism for nucleation of the stable phase following primary recalescence from undercooled melts, the role of convection in phase selection may be evaluated. The value of this research is that the results have application to the design of industrial welding, spray forming and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection and analysis of the processes governing microstructural		

	evolution. Experiments focus on the use of convection as a processing control parameter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	Highlights of Work Accomplished During Previous Grant Period 2009-10. • Attended ThermoLab working group for sample selection Friedrichshafen 4/15/09 • Initiated flight planning for LODESTARS steel sample with Fecht/Wunderlicht • Initiated flight planning for shared MAGNEPHAS sample with Loser/Volkmann • Organized 17th Symposium on Thermophysical Properties, Boulder CO 6/21/09 • Attended 1st MSL-EML Working Group at ESA/ESTEC 7/2/09 (B-1 samples) • Attended 2nd MSL-EML Working Group at ESA/ESTEC 12/18/09 (B-2 samples) • Hired Tufts PhD student Paul Choufani to conduct flight support testing at MSFC ; • Hired Tufts MS student Matthew Sherman to conduct flight support testing • Conducted ESL baseline tests on FeCo alloys at MSFC 1/11-15/2010 - Tested 10 samples of 50%Fe-50%Co alloy and 10 samples of 45%Fe-55%Co alloy corresponding to flight alloy compositions to be run on the International Space Station as part of ESA MAGNEPHAS collaboration. - Samples yielded 37 double recalescence cycles and 7 single recalescence cycles with undercoolings ranging from 15 to 152 degrees - Conducted emissivity evaluations on 3 samples of 80%Fe-20%Co alloy. • Submitted proposals for expanded collaboration under European Science Foundation AO-2009-1020 for MSL-EML ISS space experiments • ESA proposals passed science peer review and are now awaiting funding: - Peritectic Alloy Rapid Solidification with EM Convection ; - Thermophysical Properties of Liquid Metallic Alloys • Became an invited member ESA International Topical Team on Solidification (SOL-EML)
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Egry I, Löhofer G, Matson D, Wunderlich R. "EML processing measurement techniques." High Temperature Materials and Processes, 2008;27(6):389-400,. , Oct-2008
Articles in Peer-reviewed Journals	Hyers RW, Rogers JR. "A review of electrostatic levitation for materials research." High Temperature Materials and Processes, 2008;27(6): 461-74. , Oct-2008
Papers from Meeting Proceedings	Hyers RW. "Anomalies in the thermophysical properties of undercool glass-forming alloys" 17th Symposium on Thermophysical Properties, Boulder CO, June 22, 2009. NIST 2009. 17th Symposium on Thermophysical Properties, Boulder CO, June 22, 2009. Proceedings. 17th Symposium on Thermophysical Properties, Boulder CO, June 22, 2009. , Jun-2009